Parth Ganeriwala, Ph.D.

RESEARCH SCHOLAR

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Summary

Ph.D. in Computer Science (Florida Institute of Technology, 2026) researching formal verification and assurance of learning-enabled, safety-critical autonomous systems, spanning transfer learning, cognitive-architecture verification, and computer vision for aviation. Author of 34 publications (30 accepted, including NASA Formal Methods, IEEE SysCon, AIAA, and CVPR/ICCV workshops) supported by NASA-, DARPA-, and NAVAIR-funded research. Advises graduate and undergraduate researchers on transfer learning and assurance frameworks for autonomous aircraft systems, and serves as an invited reviewer for NeurIPS, ICML, and ICLR.

Education

Florida Institute of Technology — *Melbourne, Florida*
PH.D. IN COMPUTER SCIENCE: GPA - 4.0/4.0 — *January 2023 – May 2026*

- Research Interests: Formal Methods, Artificial Intelligence, Machine Learning, Deep and Transfer Learning, LLMs and Automation
- *Dissertation:* **What is Common Knowledge Across Domains? Investigating Shared Representations in Transfer Learning** *(Advisor: Dr. Siddhartha Bhattacharyya)*
- Honors: **Outstanding Student of the Year** (Honors Convocation 2025); Inducted Member of **Phi Kappa Phi**

Florida Institute of Technology — *Melbourne, Florida*
M.S. IN COMPUTER SCIENCE: GPA - 4.0/4.0 — *May 2022 – December 2023*

- Core subjects: Artificial Intelligence, Database Systems, Formal Methods, Advanced Software Engineering, Speech Recognition.

Birla Institute of Technology and Science, Pilani — *Dubai, United Arab Emirates*
B.E. IN COMPUTER SCIENCE: GPA - 3.5/4.0 — *August 2018 – December 2021*

- Thesis: *ERF-CondLaneNet: an Ego Car Lane Detection Framework*

Publications

Why Trust Machine Learning? Evolution of Trustworthiness Debt Leveraging Trustworthiness Attributes — *Under Review*
KMAL FARABI, A PRITCHETT, M SYNBORSKI, J LIU, M CARROLL, M NAMUKASA, N MANNING, R SEN, MAH KHAN, **P GANERIWALA**, S BHATTACHARYYA, S RAY, S BHUNIA, M MAHMOUD, A HOMAIFAR, P KEBRIA, M PAVONE, M GANAI AND C LIU — *AAAI 2027*

It Is Not the Model, It Is the Input: Test-Time Input-Space Refinement for Few-Shot Segmentation — *Under Review*
MAH KHAN, **P GANERIWALA**, A ALVAREZ AND S BHATTACHARYYA — *AAAI 2027*

Surveying the Landscape of Transfer Learning: Common Knowledge and Beyond — *Under Review*
P GANERIWALA, N NEOGI AND S BHATTACHARYYA — *IEEE Trans on Pattern Analysis and Machine Intelligence*

Formalizing Natural Language Specifications for MBSE with an NLP-Based Grammar for Automated System Architecture Generation — *Under Review*
C CHAMBERS, **P GANERIWALA**, S MUELLER, S BHATTACHARYYA AND C SEN — *IEEE Systems Journal*

VIROPLEX: A GNN-Driven Assessment of the Pandemic Potential of Zoonotic Bat Viruses — *Accepted*
A AGNIHOTHRAM, **P GANERIWALA**, AND S BHATTACHARYYA — *IEEE ICTAI 2026*

Formal Analysis of Stochastic Cognitive Models: A Translation Framework from Soar to PRISM — *Accepted*
P GANERIWALA, C CHAMBERS, JI LATHROP, S BHATTACHARYYA, J BABAR, S GILBERT, I AMUNDSON, M DORNEICH AND MAH KHAN — *NASA Formal Methods 2026*

Compositional Reasoning over System Architectures with Integrated Cognitive Model — *Accepted*
P GANERIWALA, C CHAMBERS, S BHATTACHARYYA, I AMUNDSON, AND J BABAR — *IEEE SysCon 2026*

Correlation-Threshold Feature Selection for Internet of Things (IoT) Device Identification under Domain Shift — *Accepted*
A DHANAWADE, **P GANERIWALA**, AND S BHATTACHARYYA — *ISAIM 2026*

AssistTaxi-v2: A Scalable Dataset for Taxiway/Runway Scene Understanding Under Diverse Conditions — *Accepted*

P GANERIWALA, MAH KHAN, A ALVAREZ, S BHATTACHARYYA, N NEOGI AND S LEHMAN — *AIAA SciTech 2026*

Explainable Assurance through Compositional Verification with Cognitive Models — *Accepted*

P GANERIWALA, C CHAMBERS, S BHATTACHARYYA, AND J BABAR — *IEEE RTSS/ESRA 2025*

Evaluating LLM Translation for Prompt-Enhanced ACT-R and Soar Models — *Accepted*

P GANERIWALA, S WU, S BHATTACHARYYA AND F RITTER — *BRIMS 2025*

Enabling Formal Verification in a Common Model of Cognition — *Accepted*

P GANERIWALA, M MATSUMURO, F RITTER AND S BHATTACHARYYA — *BRIMS 2025*

Integrating Reconfigurable Accelerators with Quantum Computing — *Accepted*

PRATIBHA, **P GANERIWALA** AND N MAHMUD — *IEEE QCE QCore Workshop 2025*

Adapt, But Don't Forget: Fine-Tuning and Contrastive Routing for Lane Detection under Distribution Shift — *Accepted*

MAH KHAN, **P GANERIWALA**, S LEHMAN, A ALVAREZ, S BHATTACHARYYA AND N NEOGI — *ICCV 2COOOL Workshop 2025*

AI Driven Differentiation and Quantification of Metal Ions Using ITIES Electrochemical Sensors — *Accepted*

M AHMED, **P GANERIWALA**, A SAVVIDOU, N BREEN, S BHATTACHARYYA, P PATHIRATHNA — *Journal of Sensor and Actuator Networks 2025*

FLAIR: Few-Shot Learning for Grapheme Recognition in Ancient Scripts — *Accepted*

P GANERIWALA AND D MITRA — *CVPR SINT4CH Workshop 2025*

Few-Shot Learning for Grapheme Recognition in Ancient Scripts — *Accepted*

P GANERIWALA AND D MITRA — *ACM Journal on Computing and Cultural Heritage 2025*

Can Someone Prove Your Operator Won't Get Distracted? A Gentle Introduction to Formal Methods in Human Factors — *Accepted*

S GILBERT, **P GANERIWALA**, J LATHROP, A NEWENDORP, S FIEFFER, P WU, I AMUNDSON, C CHAMBERS, A KOHL, S KHAN, M SANAEI, J BABAR, T WANG, D MUSLINER, R GOLDMAN, J GOTTLIEB, E WINER, M DORNEICH AND S BHATTACHARYYA — *HFES 2025*

Modeling and Formal Analysis of High-Assurance Mixed-Reality Systems — *Accepted*

I AMUNDSON, J BABAR, H HERENCIA-ZAPANA, S F ROLLINI, B BRUSSEE, P WU, T WANG, D MUSLINER, R GOLDMAN, J GOTTLIEB, A NEWENDORP, A KOHL, S FIEFFER, S KHAN, M SANAEI, M MUSCALA, S GILBERT, E WINER, M DORNEICH, J LATHROP, **P GANERIWALA**, C CHAMBERS AND S BHATTACHARYYA — *AIAA DATC/IEEE DASC 2025*

Systems Engineering with Architecture Modeling, Formal Verification and Human Interactions for Learning-Enabled Autonomous Agent — *Accepted*

P GANERIWALA, R JONES, M MATESSA, S BHATTACHARYYA, J DAVIS, S ROLLINI, H PUROHIT, N NEOGI — *INCOSE Systems Journal*

Design and Validation of Learning Aware HMI for Learning-Enabled Increasingly Autonomous Systems — *Accepted*

P GANERIWALA, M MATESSA, S BHATTACHARYYA, R JONES, J DAVIS, P KAUR, S ROLLINI, N NEOGI — *IEEE SysCon 2025*

Automating Physics-Based Reasoning for SysML Model Validation — *Accepted*

C CHAMBERS, S MUELLER, **P GANERIWALA**, S BHATTACHARYYA AND C SEN — *IEEE SysCon 2025*

Runway vs. Taxiway: Challenges in Automated Line Identification and Notation Approaches — *Accepted*

P GANERIWALA, A ALVAREZ, A ALQAHTANI, S BHATTACHARYYA, MAH KHAN, N NEOGI — *IEEE SysCon 2025*

Exploring Machine Learning Engineering for Object Detection and Tracking by Unmanned Aerial Vehicle (UAV) — *Accepted*

A GUNA, **P GANERIWALA**, AND S BHATTACHARYYA — *IEEE ICMLA 2024*

ALINA: Advanced Line Identification and Notation Algorithm — *Accepted*

MAH KHAN, **P GANERIWALA**, S BHATTACHARYYA, N NEOGI AND R MUTHALAGU — *CVPR VDU Workshop 2024*

AssistTaxi: A Comprehensive Dataset for Taxiway Analysis and Autonomous Operations — *Accepted*

P GANERIWALA, S BHATTACHARYYA, S GUNTHER, B KISH, MAH KHAN, A DHADOTI AND N NEOGI — *IEEE ICMLA 2023*

Towards Knowledge Extraction and Parsing of XML Metadata for SysML System Architecture Modeling — *Accepted*

C CHAMBERS, **P GANERIWALA**, S BHATTACHARYYA, C SEN AND N NUR — *IEEE UEMCON 2023*

Automated Framework to Extract Software Requirements from Source Code — *Accepted*
C MISKELL, R DIAZ, **P GANERIWALA**, K SLHOUB, F NEMBHARD — *ACM NLPIR 2023*

Assuring Learning-Enabled Increasingly Autonomous Systems (ALEIAS) — *Accepted*
N NARAYAN, **P GANERIWALA**, R JONES, M MATESSA, S BHATTACHARYYA, J DAVIS, H PUROHIT AND S ROLLINI — *IEEE SysCon 2023*

IPAssess: A Protocol-Based Fingerprinting Model for Device Identification in IoT — *Accepted*
P GANERIWALA, S NANDANWAR, A GUPTA, S BHATTACHARYYA AND R MUTHALAGU — *SAI IntelliSys 2023*

Cross Dataset Analysis and Network Architecture Repair for Autonomous Car Lane Detection — *Accepted*
P GANERIWALA, S BHATTACHARYYA AND R MUTHALAGU — *IEEE Intelligent Vehicles Symposium (IV) 2023*

Functional Reasoning of System Architecture in the System Modeling Language (SysML) With XML Representation — *Accepted*
P GANERIWALA, C CHAMBERS, C SEN AND S BHATTACHARYYA — *ASME IDETC-CIE 2023*

Modeling IoT Behavior for Enforcing Security and Privacy Policies — *Accepted*
A GUPTA, D CAMPOS, A DCOSTA, **P GANERIWALA**, S BHATTACHARYYA AND T OCONNOR — *SAI Computing Conference 2022*

Toward Generating System Architecture and Formal Functional Description in the Architecture Analysis & Design Language (AADL) With Structured Natural Language — *Accepted*
A CHAUHAN, **P GANERIWALA**, C SEN AND S BHATTACHARYYA — *ASME IDETC-CIE 2022*

Experience

Software and Systems Engineering, Avidyne — *Melbourne, FL*
SOFTWARE ENGINEER II — *May 2026 – Present*

- Design and implement **embedded C++ software** for avionics flight display systems, spanning **operating system services**, **device drivers**, and **touchscreen-based user interfaces**.
- Verify embedded software correctness through **unit and integration testing**, supporting **DO-178C** and **FAA** certification audits.
- Led testing for the **Vantage 12 major software release** (**60 tests across 2 hardware configurations**), cutting the release cycle by **67%** compared with the previous minor release.
- Co-create an internal **Claude Code toolkit** for **DO-178C** certification work (reusable skills for requirements lookup, traceability audits, and AI-graded test logs), adopted by **5 engineers**; built an AI test-log grading workflow targeting ∼**20 minutes** of manual review per run.
- Mentor extbf6 early-career engineers, including extbf4 interns, guiding onboarding and technical growth.

Software and Systems Engineering, Avidyne — *Melbourne, FL*
SOFTWARE ENGINEER INTERN / CO-OP — *May 2025 – May 2026*

- Expanded **end-to-end (E2E) test and evaluation (E2TE)** coverage across navigation, communication, flight management, and flight display systems, increasing overall test coverage by **25%**.
- Developed, integrated, and debugged **C-based flight software** in the avionics stack, supporting compliance with real-time, safety-critical **DO-178C** guidelines for production-ready systems.
- Built **30+ system-level and flight-specific test cases** and **TFC (Test For Credit)** software tests, reducing initial verification cycle time by **10%**.
- Merged product test libraries across multiple avionics platforms, creating unified cross-product testing infrastructure that improved code reusability and maintenance efficiency by ∼**30%**.
- Enhanced automation capabilities by developing C and Visual Basic test scripts for complex regression and validation processes, accelerating simulation turnaround by 20% and enabling seamless cross-product test execution.
- Executed hardware-in-the-loop simulations using proprietary avionics simulation frameworks, diagnosing and resolving complex execution issues to improve simulation-to-aircraft fidelity by ∼**10%**.
- Supported flight code development for Avidyne's Quantum Open Avionics Platform, supporting rapid prototyping of next-generation avionics solutions and enabling faster deployment cycles.
- Automated initial regression processes by writing Visual Basic test scripts, improving basic simulation validation efficiency by ∼**15%**.

ASSIST Research Lab, Florida Institute of Technology — *Melbourne, FL*
RESEARCH SCHOLAR — *August 2021 - Present*

- Working with a team of research professionals for formal methods of verification and run-time assurance, ML, IoT, robotics, and cyber security.
- Helped develop **research proposals with professors** on diverse topics, including cognitive agents on human behavior, the assurance of artificial intelligence in safety-critical systems, and the fine-tuning of LLMs for domain-specific queries.
- Collaborating on NASA's ***University Leadership Initiative (Round 8)*** as part of a **Florida Tech**-led multi-university/industry team to develop a framework for **trustworthy, increasingly autonomous aviation safety systems**; partners include ***Penn State, NC A&T State, UF, Stanford, Santa Fe College, Uni of New Mexico, Collins Aerospace, and ResilienX***; part of awards totaling up to $6.9M over three years.
- Modeling human cognitive behavior representation formally with respect to cyber-sickness in AR/VR systems, with ***Collins Aerospace, Iowa State University, RTX Technologies Research Center (RTRC), and Smart Information Flow Technologies (SIFT)***, funded by ***DARPA***.
- Applied **Large Language Model (LLM) translation** to cognitive architectures with ***Penn State University***, focusing on enhancing the integration of LLMs to facilitate communication and knowledge transfer within cognitive systems.
- Developing a modeling approach for designing, maintaining, and supporting air and sea platform fiber optic communications technology with ***Critical Frequency Design***, funded by ***Naval Air Systems Command***.

- Formally verifying the autonomous agent to assure safety as well as the logical correctness of the safety-critical system, with ***Rockwell Collins and Soar Tech***, funded by ***NASA***.
- Advising a Ph.D. student and undergraduate students on transfer learning, automated data labeling, and assurance frameworks for vision-based classification in autonomous aircraft systems funded by ***NASA***, addressing safety and reliability challenges in aviation technologies.
- Advised **4 groups of computer science students** on the design, development, and deployment of software for their senior projects
- Mentoring **20–30 undergraduate and high school students** on machine learning engineering approaches in the aerospace and systems domains, leading to **11 publications** that addressed real-world challenges in these fields.
- Investigated the development of a **cognitive-enhanced agent** for automatically piloting aircraft in dense urban environments which emphasized safe and reliable takeoff/landing among aerial traffic without human intervention.
- Created **AssistTaxi**, a novel dataset for runway and taxiway analysis published at IEEE ICMLA 2023 and extended as AssistTaxi-v2 (AIAA SciTech 2026), contributing to autonomous operations.
- Developed an **autonomous aircraft perception system** for accurately detecting and labeling line markings on an airport taxiway.
- Assisted with the **formulation of quizzes and homework projects** for the courses: Python, Database Systems, Web Applications, Big Data and Management, and Software Metrics.

L3Harris Institute for Assured Information, Florida Institute of Technology ***Melbourne, FL***

GRADUATE RESEARCH ASSISTANT ***May 2024 – June 2024***

- Developed a **decentralized framework** enabling **multiple autonomous agents**—robotic dogs, drones, and mobile robots—to coordinate, communicate, and reach shared goals.
- Collaborated with **developers and professors** to rigorously test the system in both **simulation** and **real-world environments**.

IRI Research, Florida Institute of Technology ***Melbourne, FL***

IRI RESEARCH DEVELOPER ***May 2023 - July 2023***

- Proposed and **implemented a framework using AI language models** to automatically extract software requirements from source code.
- Supervised and coordinated with undergraduate students towards the development process.

Program Committees

Feb 2026	**Invited Review Committee Member**, International Conference on Artificial Intelligence, Computer, Data Sciences and Applications	*Boracay Island, Philippines*
Dec 2025	**Invited Reviewer**, 24th IEEE International Conference on Machine Learning and Applications (ICMLA)	*Boca Raton, FL*
Oct 2025	**Invited Reviewer**, 18th International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation.	*Pittsburgh, PA*
Feb 2025	**Invited Reviewer: Top Reviewer**, NeurIPS 2025	*San Diego, CA*
Dec 2024	**Invited Reviewer**, 23rd IEEE International Conference on Machine Learning and Applications (ICMLA)	*Miami, FL*
Dec 2024	**Invited Reviewer**, International Conference on Electrical and Computer Engineering Research	*Gaborone, Botswana*
Dec 2024	**Invited Reviewer**, International Conference on Machine Learning (ICML) 2025	*Vancouver, Canada*
Nov 2024	**Invited Reviewer**, The 4th International Conference on Electrical, Computer, Communications and Mechatronics Engineering	*Male, Maldives*
Sep 2024	**Invited Reviewer**, International Conference on Learning Representations (ICLR) 2025	*Singapore EXPO*
Aug 2024	**Invited Reviewer**, 23rd IEEE International Conference on Machine Learning and Applications	*Miami, FL*
Jun 2024	**Invited Reviewer**, International Conference on AI, Information Processing and Cloud Computing	*Sanya, China*
May 2024	**Invited Reviewer**, NeurIPS 2024	*Vancouver, Canada*
Aug 2023	**Invited Reviewer**, 22nd IEEE International Conference on Machine Learning and Applications (ICMLA)	*Jacksonville, FL*
Nov 2022	**Invited Reviewer**, 2023 American Control Conference (ACC)	*San Diego, CA*

Skills

Programming	Java, C/C++, Python, Visual Basic, Bash, MATLAB, LaTeX
Data Analytics	Jupyter, pandas, Dask, Statsmodels, Seaborn, MySQL, MongoDB, MySQL Workbench, Neo4j, Elasticsearch
Machine Learning Libraries	scikit-learn, XGBoost, OpenCV, nltk, pandas, scipy
Deep Learning / LLMs	TensorFlow, PyTorch, Keras, CUDA, TensorRT, BERT, GPT, Hugging Face Transformers, Hugging Face Accelerate, PEFT/LoRA, LangChain, LlamaIndex, Claude Code
Cloud & MLOps	AWS (S3, SageMaker, EC2), Ray, Weights & Biases, MLflow
Formal Verification	NuSMV/nuXmv, Uppaal, AGREE, PRISM, TLA+ - Coq (Class Projects)
Robotics & Systems Engineering	Robot Operating System (ROS), AADL, SysML, Gazebo, RViz, Agile Software Development
Other Tools	Docker, Git, Perforce, GitHub Actions, NLP Modules, SparkAR Studio, Android Studio, Arduino